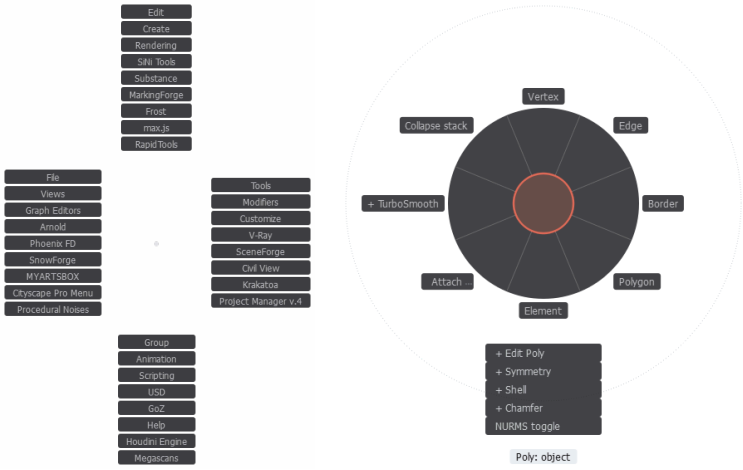
# MarkingForge

for Maya users - your marking menus and hotbox, in 3ds Max

3DS MAX 2027 · VERSION 1.5.122




## REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears when you hold the key with the mouse still - after a tenth of a second - to remind you where things are. After a few days your hand knows the directions and you stop looking.

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If your hand already knows Maya's marking menus, it already knows MarkingForge: hold, aim, release. This guide maps what you know from Maya to where it lives here, and sets 3ds Max up the way a Maya user expects in five minutes.

# Contents

New since 1.5.70 - read this first

1. You already know how this works
2. Maya habit - MarkingForge equivalent
3. Set 3ds Max up like Maya in five minutes
4. Differences worth knowing
5. A two-week plan for a Maya hand
6. Support, bug reports and feature requests

# New since 1.5.70 - read this first

What changed how MarkingForge feels - each takes a second to try. Questions and ideas: the MarkingForge Discord, **https://discord.gg/HsakZWQZAw**.

### You never have to hit a tile to confirm

A dial picks by the **direction** you move in, not by the caption under the cursor. Move towards a command and let go anywhere that way - near the centre, on its caption or far past it: the command runs. Only the **centre** cancels. In Maya's shape the caption under the cursor wins.

### The dial opens in a tenth of a second (1.5.72)

Hold the key with the mouse still: the dial is there after 100 ms - 3ds Max alone waits 250 ms. **Dial speed...** in the editor sets 0 to 250 ms and an opening animation. A flick - moving at once - never waits: the command runs and no dial is drawn.

### Your size (1.5.70, 1.5.73)

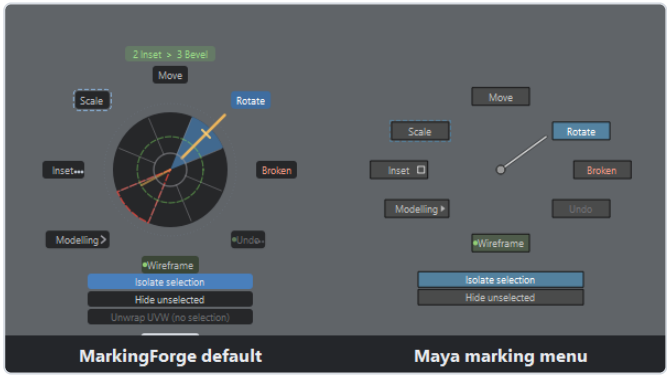
**Dial size...** in the editor: the whole dial 60 to 160 %, the text 70 to 200 %, and the green return ring of multiple picks - with a picture of the real dial while you drag.

### Any key - Tab too (1.5.73, 1.5.75)

A dial can sit on any key. On Tab it opens from the viewport and the panels alike and never walks the keyboard into a number field. Ctrl+Z silent while the toolbar's Undo works? The keyboard is in a field: press Esc.

### Exactly Maya's marking menu (1.5.72 - 1.5.74)

One tick: **MarkingForge > Experimental features > Maya marking menu look**. Every dial is then drawn as in Maya: no disc, grey boxes round a small centre dot, a line to the cursor, Maya's blue on the box you aim at, and Maya's **option boxes** - let go on the small square beside a command to open its settings. Clear the tick and your own colours are back. For one dial only: **Colours...** > preset *Maya marking menu*, or just its **Shape** list.




The same dial in both shapes: the MarkingForge dial (left) and the Maya marking menu look (right) - Maya's option box beside *Inset*.

### Newest (1.5.98 - 1.5.117)

**Chain submenus**: a submenu opens under the cursor, clear of the small circle you slide back onto - **Chain distance...** in the editor opens it farther (40 px by default). **Your own layout** (beta): drag a dial's boxes in the editor's drawing or in **Layout...**; **Separator above...** puts a line or a gap between the rows under the dial. **Pivots the Maya way**: the Extra dial *Pivot (Maya style)*. The **selected object wins** under the cursor. **Your dials in a scene**: Experimental features > *Store all my dials in this scene*.

# You already know how this works

MarkingForge brings the interaction you use in Maya every day into 3ds Max, the same way round:

**IN MAYA**

Hold a key or a mouse button, a radial menu appears, drag towards an item, release. Experienced users flick without looking at the menu.

→

**IN MARKINGFORGE**

Exactly that. Hold a key, the dial appears, aim, release - or flick without waiting for the dial. Release in the centre to cancel.

**IN MAYA**

Eight compass positions, plus extra items in a list under the marking menu; sub-menus open by dragging through an item.

→

**IN MARKINGFORGE**

Eight directions, plus up to twelve rows under the dial; submenus open by moving through their direction, four levels under the dial.

**IN MAYA**

The marking menu has no disc: grey boxes round a small centre dot, a line to the cursor, and it appears as soon as the mouse is still.

→

**IN MARKINGFORGE**

The same look in one tick (1.5.74):
**MARKINGFORGE > EXPERIMENTAL FEATURES > MAYA MARKING MENU LOOK**

- or per dial:
**COLOURS...**

> preset *Maya marking menu*, option boxes included (1.5.73).
**DIAL SIZE...**

at 60 % gives Maya's size and sets the text size; the dial opens 100 ms after the key for a still hand, and
**DIAL SPEED...**

at 0 opens it at once. Flicks never wait.

**IN MAYA**

The right-click marking menu changes with what is under the cursor and with the component mode.

→

**IN MARKINGFORGE**

Context variants: the same key shows different tools at vertex, edge or polygon level, on a spline, over the material editor - automatically.

**IN MAYA**

Hold Space: the hotbox shows every menu at the cursor.

→

**IN MARKINGFORGE**

The hotbox dial shows every 3ds Max menu as tiles at the cursor - including menus other plugins add. Put it on Space if you like (chapter 3).

**IN MAYA**

More marking menus from the same place: each zone of the hotbox has its own marking menus, one per mouse button.

→

**IN MARKINGFORGE**

One key can hold several sets of a dial - its pages. Hold the key and turn the mouse wheel: the next or previous page appears in place

("page 2 of 3" under the dial). The next press opens the page you turned to last.

# Maya habit - MarkingForge equivalent

| In Maya you... | In MarkingForge | Notes |
|---|---|---|
| **Hold Space for the hotbox** | Alt+5, one mouse-wheel notch away (the hotbox page) | Or put the hotbox dial from the dial library on a key of its own - Space, chapter 3. |
| **Right-click an object to pick a component mode** | Alt+1 on an Editable Poly at object level: Vertex N, Edge NE, Border E, Polygon SE, Element S | 3ds Max's own keys 1-5 still work. |
| **Shift + right-click for the modelling tools of the current component** | Alt+1 at a sub-object level: vertex / edge / border / polygon / element tools | Same directions on Editable Poly and under Edit Poly - except W at polygon level (Detach / Bridge). NW goes back to object level. |
| **Ctrl + right-click to convert the selection** | Alt+3 at a sub-object level of an Editable Poly: To vertex, To edge, To face on three corners | Grow / Shrink and Loop / Ring are on the same dial. |
| **Grow / shrink the component selection** | Alt+3 at a sub-object level: Grow N, Shrink S | |
| **Click an item's option box to set it up first** | With the Maya marking menu look (Colours...): release on the small square beside the command, exactly as in Maya (1.5.73). With the MarkingForge dial: move far past the ring before releasing - a long flick counts as far | Per item: Settings... in the editor chooses which is the default. |
| **Repeat the last command (Maya's G)** | A tap of any dial's key, with Experimental features > Tap repeats the last modelling command ticked (1.5.85): your last poly, mesh or spline command again. Or Alt+8 (Recent commands), north is the newest | Each kind of object remembers its own last command. |
| **Undo several steps** | Alt+7 (Undo): each direction undoes more steps, by name | N = 1 step, NE = 2 ... |
| **Frame the selection / frame all** | Ctrl+Alt+7: Zoom selected SW, Zoom extents all N | 3ds Max's Z key also frames the selection. |
| **Isolate the selection** | Ctrl+Alt+7: Isolate S | The same move ends the isolation. |
| **Switch between perspective and orthographic views** | Ctrl+Alt+8 (Viewports): a compass of views | Top N, Bottom S, Left W, Right E, Front SE, Back SW, Perspective NE, Camera NW. |
| **Create primitives from a marking menu** | Alt+4 (Create) | Shapes when a shape is selected. |
| **Reach several marking menus from one place (the hotbox zones)** | Sets of one dial: hold its key and turn the mouse wheel | Add a set in the editor with **New set...** or **Dial library...** > **Add as a new set of Menu N**; each set has a tab over the dial's contents. |
| **Build your own marking menu in the Marking Menu Editor** | MarkingForge > Menu editor... | Any of 4000+ 3ds Max commands, 128 ready-made scripts, your own scripts, submenus. Or start from **Dial library...** or **New dial wizard...** |
| **Take your marking menus to another machine** | Right-click a dial in the editor: save it to a .mfdial file, or **Save every dial...** | On the other machine: **Load dial...** or **Load every dial...**. Scripts from somebody else's files are removed unless you keep them. |

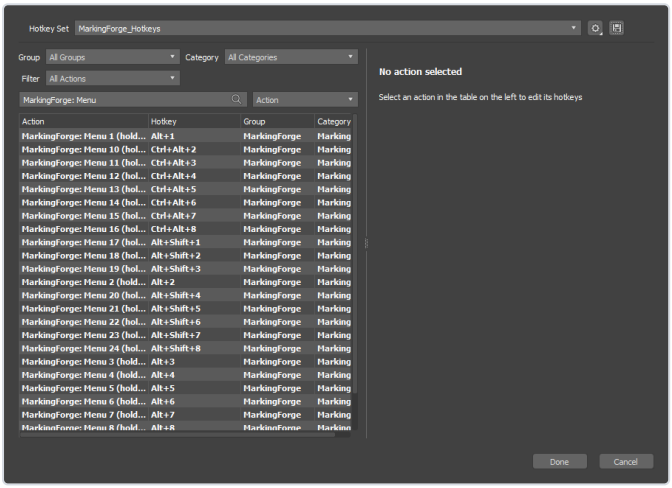
| | | |
|---|---|---|
| **Give a marking menu a hotkey (press / release)** | Set shortcut... + Apply shortcuts | Or 3ds Max's Customize > Hotkey Editor: MarkingForge: Menu 1-24. |
| **Put a marking menu on a mouse button** | Mouse button (+ Ctrl / Alt / Shift) in the editor | A modifier is required - chapter 4 explains why. |

# Set 3ds Max up like Maya in five minutes

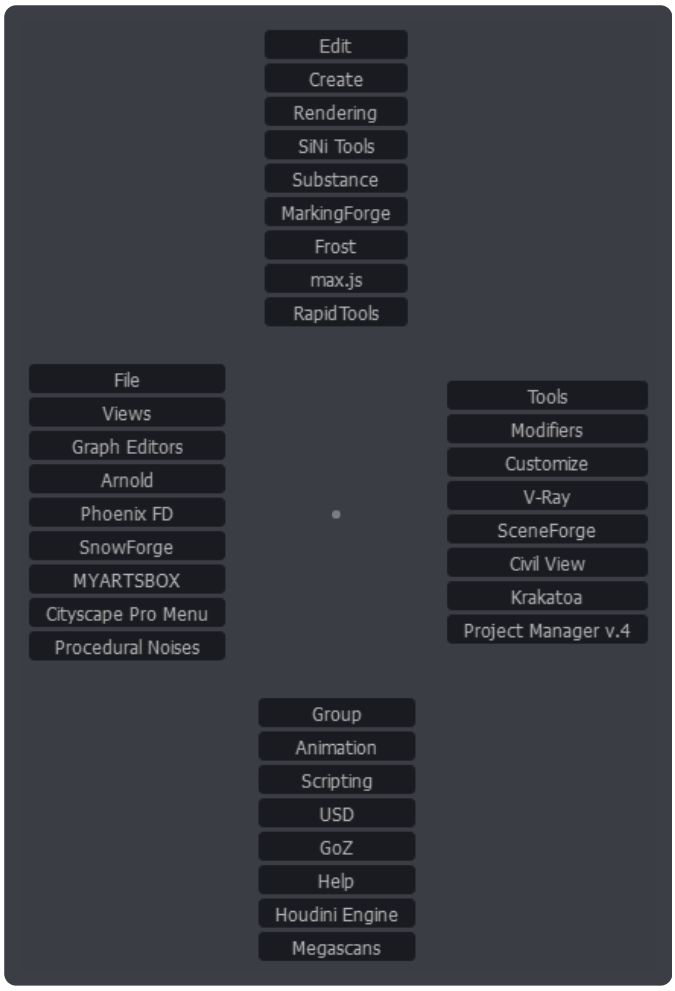
## 1. The hotbox on Space

In 3ds Max, Space toggles the selection lock. If you do not use that, give Space to the hotbox - the selection lock stays one gesture away on Alt+3 (west).

1. **Customize > Hotkey Editor**.
2. Type `MarkingForge: Menu 5` in the search field and select it.
3. Click the hotkey field on the right, press **Space** and assign it. 3ds Max tells you Space is used by Selection Lock Toggle - reassign.
4. **Done**. Hold Space in a viewport and turn the mouse wheel one notch away from you: the hotbox.




The Hotkey Editor filtered to the MarkingForge dials.



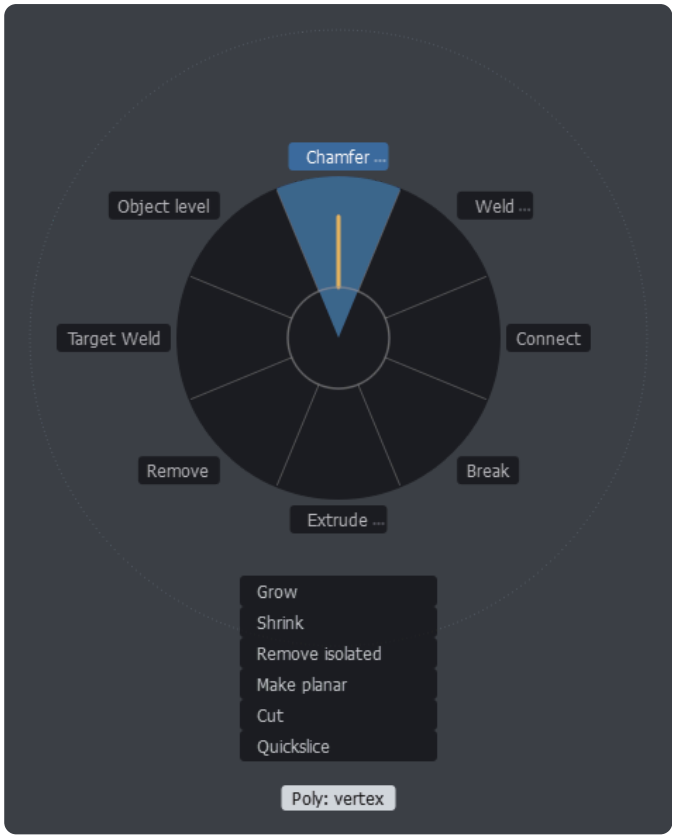

The hotbox - 3ds Max's menus around the cursor.

## 2. Component modes and tools under one key

Nothing to set up - Alt+1 already behaves like Maya's right-click and Shift + right-click menus combined. On an Editable Poly at object level it offers the component levels; at a level it offers that level's tools.

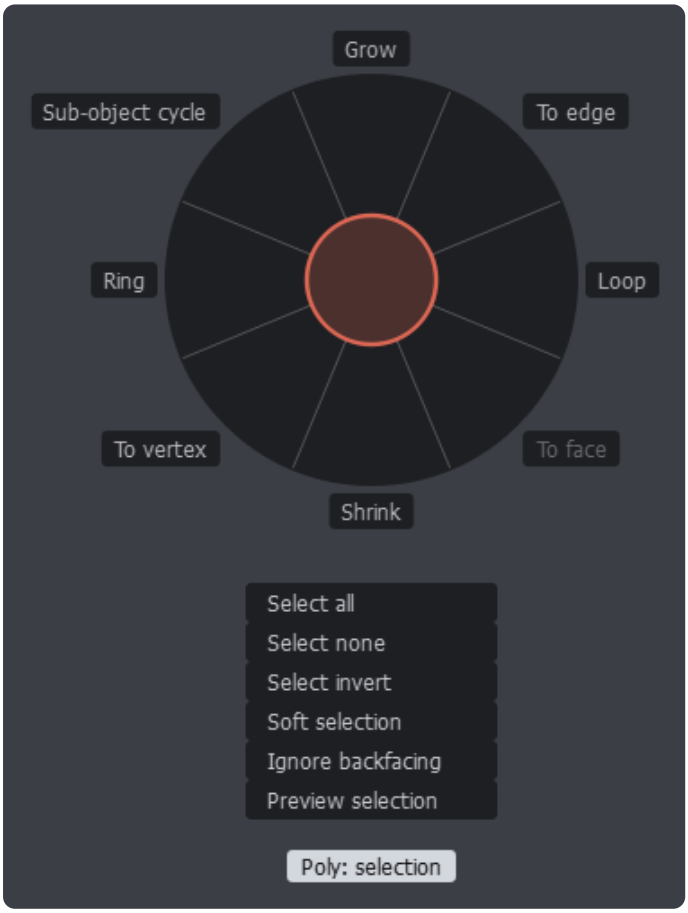



Alt+1 at object level: pick the component mode.




Alt+1 at vertex level: the vertex tools.

## 3. Selection conversion and growing

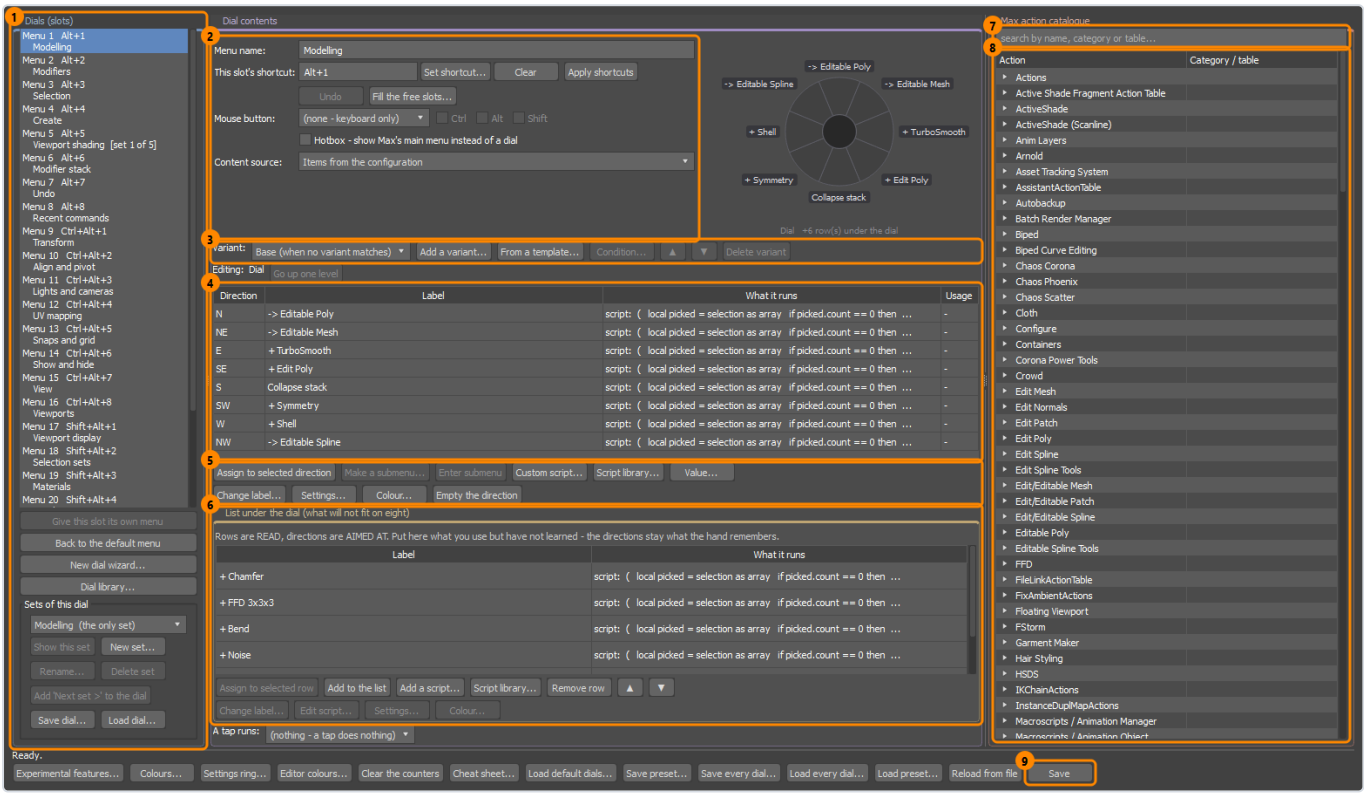



Alt+3 at a component level: Grow / Shrink, Loop / Ring, conversions on the corners.

## 4. Your most used Maya menu, rebuilt

Open **MarkingForge > Menu editor...**, pick an empty or rarely used dial and place your commands. The QuickStart Long walks through it step by step (lessons 8-13).

Want to keep the dial that is there? Make your menu a **new set** of it instead (**New set...** under **Sets of this dial**): the key still opens what it opened, and the mouse wheel turns to your menu while you hold the key. **Put this set on the key** swaps them when your menu has earned the first place. The QuickStart Long covers sets and the dial library in lessons 20 and 21.




The menu editor: dials, directions, and the catalogue of every 3ds Max command.

## 5. Your own arrangement (beta)

Maya always puts the boxes in the same eight places. MarkingForge lets you move them (1.5.107): in the editor, **Layout... (beta)** under the directions shows the menu's boxes - drag them where your hand expects them, move them with the arrows a few pixels at a time, or start from *six - two on each side*. They may even stand a little further out than Maya's own (1.5.108). Leave a direction empty and its place stays free; a flick towards an empty place does nothing, as in Maya. Long list under the menu? **Separator above...** groups it with a line or a gap, like Maya's dividers.

## 6. Pivots the Maya way

Put the extra dial **Pivot (Maya style)** on a key (**Dial library...** in the editor; 1.5.112). A dial opens where you point, so what Maya does with D and a click is one move here: open the dial over the spot and pick the direction.

| In Maya | On the dial | What it does |
|---|---|---|
| **D / Insert - edit the pivot** | W: **Edit pivot** | 3ds Max's Affect Pivot Only, on and off - move and rotate the pivot with the usual tools. |
| **D + click a face** | N: **Pivot here** | The pivot on the spot you opened the dial over. |
| **D + click, with orientation** | NE: **Pivot here + normal** | The same, with the pivot's Z along the surface there. |

| | | |
|---|---|---|
| **D + V onto a vertex** | SW: **Pivot to vertex here** | Onto the nearest corner of the polygon there. |
| **Center Pivot on components** | E: **Pivot to selected parts** | The centre of the selected vertices, edges or polygons (Editable Poly or Edit Poly). |
| **Component axis orientation** | SE: **Pivot to parts + normal** | The same, with Z along the selected polygons' average normal. |
| **Match Transformations > Match Pivots** | NW: **Match pivot** | Every selected object takes the pivot of the one selected last. |
| **Pin Component Pivot** | the list: **Copy pivot** / **Paste pivot** | Remember a pivot and give it to other objects. |
| **Center / Zero Pivot, bottom, sides** | S: **Bounds** > and the list | Top, bottom, each side, both corners of the box; the centre; the world's axes; 0,0,0. |
| **Reset (Customize Pivot)** | the list: **Reset pivot** | Back to the object's own origin and axes - the same as 3ds Max's Reset Pivot (1.5.122). |
| **Pivot orientation: object / world** | the list: **Pivot axes to object**, **Pivot aligned to world** | The pivot turned to the mesh's own axes or to the world's, left where it is. |
| **Component orientation on an edge** | the list: **Pivot along edge** | The middle of the selected edges, X along them, Z along their polygons - a spline's segments too. |
| **Match Pivots, place or turn only** | the list: **Match pivot place**, **Match pivot axes** | Only where the last one's pivot is, or only how it is turned. |
| **D + X onto the grid** | the list: **Pivot to grid** | The nearest point of the home grid. |
| **One pivot for several objects** | the list: **Pivot to selection** | Every selected object's pivot on the centre of the whole selection. |

Maya's snapping while the pivot is edited has a dial of its own, **Pivot snapping (Maya style)** (1.5.122) - again, open it over the spot:

| In Maya | On the dial | What it does |
|---|---|---|
| **D + click a face** | N: **Pivot here** | The pivot on the spot you opened the dial over. |
| **D + V onto a vertex** | NE: **Pivot to vertex here** | Onto the nearest corner of the polygon there. |
| **D + C onto an edge** | E: **Pivot to edge here** | Onto the nearest point of the nearest edge you see there - not a diagonal hidden inside a quad. |
| **Snap to a face's centre** | SE: **Pivot to polygon here** | The middle of the polygon there. |
| **D + X onto the grid** | S: **Pivot to grid** | The nearest point of the home grid. |
| **Ctrl+Shift click a component** | SW: **Orient pivot here** | The pivot's Z along the surface there; the pivot stays where it is. |
| **Ctrl click a component** | the list: **Aim pivot here** | The pivot's X pointed at the spot there; the pivot stays where it is. |
| **D / Insert** | W: **Edit pivot** | 3ds Max's Affect Pivot Only, on and off. |
| **D + click, with orientation** | NW: **Pivot here + normal** | Placed and turned along the surface. |

The selected parts can be an Editable Poly's, an Edit Poly's, an Editable Mesh's or a bare Editable Spline's (its knots, segments or splines). Nothing moves but the pivot: the mesh and any children stay where they are. With Auto Key on, every pivot script leaves the object alone. One that turns the pivot also refuses an

animated object and one scaled unevenly; one that only moves it refuses an object that turns or scales in its animation - an object that only moves keeps its animation. Each says why in the MAXScript Listener.

# Differences worth knowing

| Topic | What is different | Why |
|---|---|---|
| **Right-click menus** | Dials open on keys (or a mouse button WITH a modifier), not on a bare right-click. | In 3ds Max the right button opens the quad menu and every modifier + right-click combination opens another quad. Taking a bare mouse button away from 3ds Max would break it, so MarkingForge refuses. |
| **Free mouse combinations** | Good choices: a thumb button (XB1 / XB2) with Ctrl or Shift, or Ctrl+Shift + middle button. | Alt + middle orbits and Ctrl+Alt + middle zooms in 3ds Max. |
| **The hotbox** | Shows 3ds Max's main menu as tiles; you point at a menu title, then at an entry and release - or click the entry, as in Maya. | The MarkingForge hotbox has no marking-menu zones like Maya's - the 24 dials and their pages under the mouse wheel play that role. |
| **Option boxes** | With the Maya marking menu look (1.5.73): Maya's own option box, a small square beside the command - release on it to open the settings; a flick never does. With the MarkingForge dial: far past the ring opens the settings, and a flick follows the same ring. | Configurable per item with Settings... in the editor; the ring's distance with Settings ring... |
| **Submenus** | They open under the cursor as in Maya (**Chain submenus**, on by default since 1.5.99) - where you stop, or as soon as the way back is clear of the new menu's boxes (1.5.115) - a little farther on in your direction (**Chain distance...**, 1.5.117) - and every menu you came through stays as a small circle - move onto one to go back to it. Untick it in Experimental features and they open in place: the dial keeps its centre, back up is a return to it. | In place since 1.4.8 on request (the dial moving with every submenu was distracting); the chain is one tick in Experimental features, in the classic dials and Maya's look alike. |
| **Context** | Chosen automatically from the sub-object level, the selection, the modifier in the panel or the window under the cursor. | The first matching variant wins - you control the order in the editor. |
| **How many menus** | 24 dials on Alt, Ctrl+Alt and Shift+Alt with digits 1-8 - and any dial can hold several sets, turned with the mouse wheel. | The digit is the place, the modifier the bank - easy to count to without looking. |
| **Pivots** | One pivot per object, and it IS the object's transform: moving it with Affect Pivot Only is final at once - there is no Bake Pivot to do, and no separate rotate and scale pivots. | 3ds Max keeps the mesh at an offset from the pivot; a temporary pivot is the Working Pivot (the Align and pivot dial). |
| **The mouse wheel** | Taken only while a dial with sets is open and at its first level - it turns the page. In a submenu, the hotbox or any other time it zooms and scrolls as usual. | One page per notch, also on a fine wheel or a touchpad. |

> 3ds Max already uses **Q W E R** for select, move, rotate and scale - the same keys as Maya - so those habits carry over without MarkingForge.

# A two-week plan for a Maya hand

| When | Do this |
|---|---|
| **Day 1** | Hotbox on Space. Use Alt+1 for component modes and tools instead of the command panel. |
| **Days 2-4** | Alt+3 for selection conversion, Ctrl+Alt+7 for framing and isolating. Print the Shortcut Card. |
| **Days 5-7** | Flick the gestures you use most without waiting for the dial. |
| **Week 2** | Rebuild your one or two favourite Maya marking menus as MarkingForge dials in the editor - or as a second page of a dial you already use, one wheel notch away. |

Next: the **QuickStart Long** (22 lessons with pictures) - or the **QuickStart Short**, about 15 minutes - and the **Reference** (every option).

# Support, bug reports and feature requests

| | |
|---|---|
| ISSUES | github.com/Forge-Plugins/MarkingForge/issues |
| ROADMAP | trello.com/b/7gJVqG8Y/markingforge-features-roadmap |
| DISCORD | discord.gg/HsakZWQZAw (questions, tips, news - with other users and the author) |
| E-MAIL | forgeplugins@gmail.com |
| WEB | https://forgeplugins.tech (forgeplugins.tech - more plugins by the author) |

**Bug reports and feature requests** are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Quick questions and tips: the **Discord** channel (**MarkingForge > Discord community** opens it). Anything you would rather not post publicly (licence, purchase) goes to the e-mail address. Every message is read.

## Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

## Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.